

Pettis Integrability Suffices for the Commuting Spectral-Radius Integral Inequality

Candidate full answer to the concluding question in arXiv:2605.12454

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Status. Candidate full answer to the literal question, subject to human review. Pettis integrability replaces norm-strong measurability. The sharper problem under bare pointwise L_s^1 measurability remains open.

1 Source question

The source is M. Arsenović, M. Krstić, M. Milović, and S. Milošević, *Strongly Integrable Operator-Valued Functions, Generated Vector Measures and Compactness of Integrals*, arXiv:2605.12454v1 (2026) [1]. Its Theorem 4.1 assumes that X^* contains no copy of c_0 and that $A \in L_s^1(\Omega, \mu, \mathcal{B}(X))$ is a norm-strongly measurable family of mutually commuting operators, and proves

$$r\left(\int_{\Omega} A_t d\mu(t)\right) \leq \int_{\Omega} r(A_t) d\mu(t).$$

On PDF page 36, the authors ask whether norm-strong measurability can be replaced by a weaker assumption.

Here $A \in L_s^1$ already says that every orbit $t \mapsto A_t x$ is strongly measurable and Bochner integrable. The extra assumption in question is strong measurability of $t \mapsto A_t$ in the norm of $\mathcal{B}(X)$.

2 Proof intuition

Pettis integration is precisely strong enough to commute with every bounded linear scalarization. Put all values A_t into the closed unital Banach algebra they generate. Commutativity lets characters calculate spectral radius, while Pettis integration lets each character pass through the integral. The pointwise character value is bounded by the pointwise spectral radius. A possible concern—that spectra can grow in a subalgebra—vanishes for radii: the ambient and subalgebra radii use the same norms of the same powers.

The paper's original hypotheses imply Pettis integrability. Indeed, its key theorem makes the strong integral a norm-countably-additive operator measure. On countably many bounded finite-measure pieces the density is Bochner integrable; the finite variation of every scalarization then glues these pieces into a global Pettis integral.

3 The general theorem

We use the standard definition: $f : \Omega \rightarrow \mathcal{A}$ is Pettis integrable if it is weakly measurable, every $\varphi \circ f$ with $\varphi \in \mathcal{A}^*$ is integrable, and for every measurable E there is an element $I_E \in \mathcal{A}$ satisfying $\varphi(I_E) = \int_E \varphi(f(t)) d\mu(t)$ for all $\varphi \in \mathcal{A}^*$.

Corollary 4.2. *Assume X has a shrinking F. D. D., $\mathcal{A} \in L_s^1(\Omega, \mu, \mathfrak{B}(X))$ is a commuting family of compact operators. Then, we have the following inequality*

$$r\left(\int_{\Omega} \mathcal{A} d\mu\right) \leq \int_{\Omega} r(\mathcal{A}_t) d\mu(t).$$

Moreover, if $\mathcal{A}_t$ are Volterra operators, so is $\int_{\Omega} \mathcal{A} d\mu$.

Proof. By Proposition 3.2 we get that $\mathcal{A} : \Omega \rightarrow \mathfrak{B}(X)$ is strongly μ -measurable function. Furthermore, as X^* is separable (because X has a shrinking F. D. D.) it cannot contain an isomorphic copy of ℓ^∞ and thus cannot contain an isomorphic copy of $\mathfrak{c}_0$, due to [12, Corollary 6, Page 23]. Final conclusion now follows from Theorem 4.1. If $\mathcal{A}_t$ is a Volterra operator for all $t \in \Omega$, then due to proved inequality we have that the operator $\int_{\Omega} \mathcal{A} d\mu$ is Volterra too. $\square$

Concluding remarks: It is not known to the authors whether the strong μ -measurability of the family $\mathcal{A}$ in Theorem 4.1 can be replaced by a weaker assumption. We also note that Lemma 4.4 provides alternative conditions for μ -measurability of function $\Omega \ni t \mapsto r(\mathcal{A}_t) \in \mathbb{R}$.

ACKNOWLEDGEMENTS

Figure 1: Exact source context and concluding question, arXiv:2605.12454v1, PDF page 36.

Theorem 1 (Pettis commuting spectral-radius inequality). *Let $\mathcal{A}$ be a complex unital Banach algebra and let $f : \Omega \rightarrow \mathcal{A}$ be Pettis integrable. Suppose that, after deletion of one null set, the elements $f(t)$ commute pairwise. If $t \mapsto r_{\mathcal{A}}(f(t))$ is measurable, then*

$$r_{\mathcal{A}}\left(\mathbf{P} \int_{\Omega} f(t) d\mu(t)\right) \leq \int_{\Omega} r_{\mathcal{A}}(f(t)) d\mu(t),$$

with the assertion understood trivially when the right side is infinite. Without measurability, the same inequality holds with the upper integral.

Proof. Delete the common null set and let D be the closed unital subalgebra of $\mathcal{A}$ generated by $\{f(t) : t \in \Omega\}$. It is commutative. If $I = \mathbf{P} \int_{\Omega} f d\mu$, then $I \in D$: for the quotient map $q : \mathcal{A} \rightarrow \mathcal{A}/D$ and every $\psi \in (\mathcal{A}/D)^*$,

$$\psi(qI) = \int_{\Omega} \psi(qf(t)) d\mu(t) = 0,$$

so $qI = 0$.

Let χ be a character of D . Extend it by Hahn–Banach to $\tilde{\chi} \in \mathcal{A}^*$. Pettis integrability gives

$$\chi(I) = \tilde{\chi}(I) = \int_{\Omega} \tilde{\chi}(f(t)) d\mu(t) = \int_{\Omega} \chi(f(t)) d\mu(t).$$

For every $a \in D$, the spectral-radius formula and the inherited norm give

$$r_D(a) = \lim_{n \rightarrow \infty} \|a^n\|_D^{1/n} = \lim_{n \rightarrow \infty} \|a^n\|_{\mathcal{A}}^{1/n} = r_{\mathcal{A}}(a).$$

Consequently $|\chi(f(t))| \leq r_D(f(t)) = r_{\mathcal{A}}(f(t))$. Hence

$$|\chi(I)| \leq \int_{\Omega} r_{\mathcal{A}}(f(t)) d\mu(t).$$

Taking the supremum over the character space of the commutative unital algebra D and using the Gelfand spectral-radius formula yields

$$r_{\mathcal{A}}(I) = r_D(I) = \sup_{\chi} |\chi(I)| \leq \int_{\Omega} r_{\mathcal{A}}(f(t)) d\mu(t),$$

as required. $\square$

4 Application to the source question

Lemma 2 (the original assumption implies Pettis integrability). *Let X be a complex Banach space such that X^* contains no isomorphic copy of c_0 . Suppose $A \in L_s^1(\Omega, \mu, \mathcal{B}(X))$ and $A : \Omega \rightarrow \mathcal{B}(X)$ is norm-strongly measurable. Then A is Pettis integrable as a $\mathcal{B}(X)$ -valued function, and its Pettis integral over every measurable E is its strong integral $\nu_A(E)$.*

Proof. By Theorem 3.3 of the source paper [1],

$$\nu_A(E)x = \int_E A_t x d\mu(t), \quad E \in \Sigma,$$

defines a norm-countably-additive $\mathcal{B}(X)$ -valued measure. Norm-strong measurability gives a sigma-finite support. Partition that support into countably many measurable sets (D_j) , each of finite measure and with $\text{ess sup}_{D_j} \|A_t\| < \infty$. On each D_j , A is Bochner integrable; therefore, for $E \subseteq D_j$,

$$\nu_A(E) = B \int_E A_t d\mu(t).$$

Fix $\Phi \in \mathcal{B}(X)^*$ and let $\lambda = \Phi \circ \nu_A$. This is a finite countably additive complex measure, hence its total variation satisfies $|\lambda|(\Omega) < \infty$. On each D_j its density is $\Phi(A_t)$, so

$$\int_{\Omega} |\Phi(A_t)| d\mu(t) = \sum_j \int_{D_j} |\Phi(A_t)| d\mu(t) = \sum_j |\lambda|(D_j) = |\lambda|(\Omega) < \infty.$$

For arbitrary E , absolute convergence and countable additivity now give

$$\int_E \Phi(A_t) d\mu(t) = \sum_j \lambda(E \cap D_j) = \lambda(E) = \Phi(\nu_A(E)).$$

Thus all scalarizations are integrable and $\nu_A(E)$ represents them, which is exactly Pettis integrability. $\square$

Corollary 3 (affirmative answer to the literal question). *In Theorem 4.1 of the source paper, norm-strong measurability may be replaced by the weaker assumption that A is Pettis integrable as a $\mathcal{B}(X)$ -valued function, provided $t \mapsto r(A_t)$ is measurable. Under this replacement the hypothesis $X^* \not\supset c_0$ is unnecessary.*

Proof. Apply Theorem 1 to $\mathcal{A} = \mathcal{B}(X)$. The preceding lemma shows that, within the original setting, the old measurability hypothesis implies the new one. $\square$

5 Strictness of the replacement

Proposition 4. *There is a Pettis-integrable, commuting family in $L_s^1([0, 1], \mathcal{B}(H))$ that is not norm-strongly measurable, where H is a Hilbert space and H^* contains no copy of c_0 .*

Proof. Let $H = \ell_2([0, 1])$, let e_t be its canonical unit vectors, and let P_t be projection onto $\mathbb{C}e_t$. Put $A_t = I + P_t$. Every $x \in H$ has countable support, hence $P_tx = 0$ for almost every t . Thus $A \in L_s^1$ and its strong integral is I .

For $\Phi \in \mathcal{B}(H)^*$, set $a_t = \Phi(P_t)$. Given finite $F \subset [0, 1]$, choose unimodular c_t so that $c_t a_t = |a_t|$. Orthogonality gives

$$\sum_{t \in F} |a_t| = \Phi \left(\sum_{t \in F} c_t P_t \right) \leq \|\Phi\|, \quad \left\| \sum_{t \in F} c_t P_t \right\| = 1.$$

Therefore $(a_t)_{t \in [0, 1]} \in \ell_1([0, 1])$ and has countable support. It follows that $t \mapsto \Phi(A_t) = \Phi(I) + a_t$ is measurable, integrable, and has integral $\Phi(I)$. Hence A is Pettis integrable with integral I .

For $s \neq t$, however, $\|A_s - A_t\| = \|P_s - P_t\| = 1$. On every conull set the range is therefore an uncountable 1-separated set. It is not essentially separably valued, so the Pettis measurability theorem rules out norm-strong measurability. Finally, the values commute and $r(A_t) = 2$, so the example is also inside the spectral-radius setting. (As an L_s^1 equivalence class it is pointwise equivalent to the constant family I ; the strictness here is at the representative level at which the source states the extra hypothesis.) $\square$

6 Verification, limitations, and novelty

The proof uses only Pettis scalarization, Hahn–Banach extension, the Gelfand formula in a commutative Banach algebra, and Theorem 3.3 of the source paper. The key ambient/subalgebra issue was checked explicitly through the norm power formula. No numerical computation is involved.

This packet does *not* prove the inequality for every bare pointwise L_s^1 family. Without Pettis compatibility, a character of the generated operator algebra need not be strong-operator continuous, and neither measurability of $\chi(A_t)$ nor the identity $\chi(\int A) = \int \chi(A_t)$ follows. That sharper endpoint remains open.

The local run indexes and a bounded web/arXiv search on 12 August 2026 used the exact title and combinations of *Pettis integral*, *commuting family*, and *spectral radius*. No later answer or exact formulation was found. Since the source is recent and the character argument is elementary, the novelty confidence is moderate; the correctness confidence is high, subject to human review of the intended scope of the word “weaker.”

References

- [1] M. Arsenović, M. Krstić, M. Milović, and S. Milošević, *Strongly Integrable Operator-Valued Functions, Generated Vector Measures and Compactness of Integrals*, arXiv:2605.12454v1 (2026).